STAT 131: Lecture Notes 4

Winter 2024

4.1 Continuous Random Variables [DS 3.2]

4.1.1 Definition and Properties

Many random variables can take values in certain intervals. For instance, human height/weight measurements, temperature, precipitation, etc.. As mentioned previously, these are called *continuous random variables*. For these types of random variables, the probability that x will be equal to a specific value is 0. Thus, we do not calculate things like $\mathbb{P}(X = x)$ for continuous random variables, and instead calculate the probability that the random variable might be in a specific interval. Consequently, we do not work with probability mass functions as we did in the case of discrete random variables, and instead work with *probability density functions*.

Definition 4.1 (Probability Density Function)**.** The probability density function (pdf) of a continuous random variable X is a non-negative function $f(x)$, where

$$P(a \le X \le b) = \int_a^b f(x)\,dx$$

where $\int_{-\infty}^{\infty} f(x)dx = 1$. unity

How is the pdf related to the probability of a specific event? For a continuous random variable X, the probability of some event E is

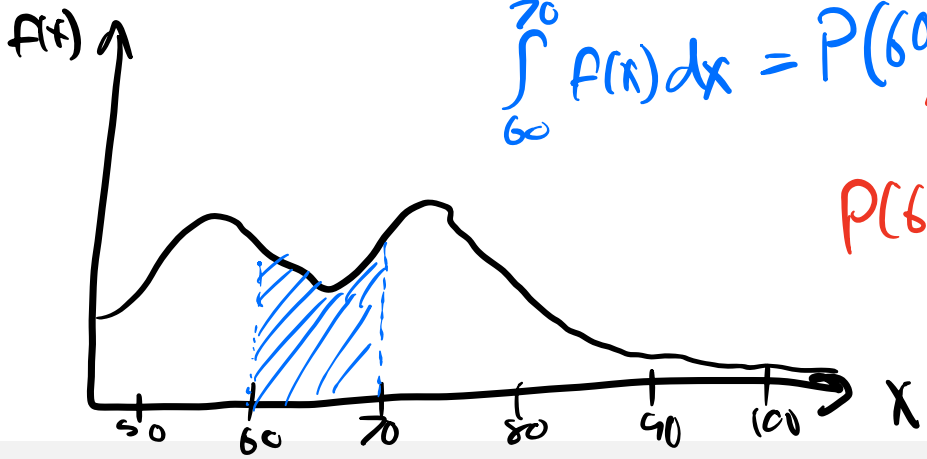
$$\mathbb{P}(E) = \int_{x \in E} f(x)dx.$$

Probability density is **not** the same as probability. For a continuous random variable X, $P(X = a) = 0$ even though we may have $f(a) > 0$.

$$\int_a^a f(x)\,dx = 0$$

It is also possible that $f(x) > 1$.

Example 4.1. PDF Diagram



temp.

$$\int_{60}^{70} f(x)\,dx = P(60 \le x \le 70)$$

$$P(60 < x < 70)$$

♢

Definition 4.2 (Support)**.** The support of a continuous random variable X is defined as the set $\{x : f(x) > 0\}$ where $f(x)$ is the pdf. In other words, it is the set of all possible values of x, where the pdf is strictly positive.

Example 4.2 (Two-year old weights)**.** Suppose that the weight of a two-year old can range from 15 pounds to 35 pounds. Let $f(x)$ denote the pdf of weight for two-year olds, where

$$f(x) = \begin{cases} \frac{3}{40} - \frac{3}{4000}(x-25)^2, & \text{for } 15 \le x \le 35 \\ 0, & \text{otherwise} \end{cases}.$$

What is the support of this random variable? What is the probability that a two-year old will weight more than 30 pounds?

support is $[15, 35]$

$$P(X > 30) = \int_{30}^{\infty} f(x)\,dx = \int_{30}^{35} f(x)\,dx$$

$$= \int_{30}^{35} \frac{3}{40} - \frac{3}{4000}(x-25)^2\,dx \approx 0.16$$

verify this

♢

Sometimes we may only know the pdf up to a constant:

$$f(x) = c \cdot g(x)$$

c is an unknown constant

where $g(x)$ is a known function and c is an unknown constant. Write may write $f(x) \propto g(x)$

Definition 4.3 (Normalizing constant)**.** For a pdf such that

$$f(x) = c \cdot g(x),$$

c is called the normalizing constant.

How can we compute c so that pdf is completely known? The idea is to use the unity property of pdf:

$$\int_{-\infty}^{\infty} f(x)dx = 1.$$

Example 4.3. For continuous random variable X, $f(x) \propto e^{-2x}$ for $x > 0$; $f(x) = 0$ for $x \leq 0$.

- Compute the normalizing constant.
- Compute $P(1 < X < 2)$.

"proportional to"
i.e. $f(x) = c \cdot g(x) = c\,e^{-2x}$

1) $1 = \int_{-\infty}^{\infty} f(x)\,dx = \int_0^{\infty} c\,e^{-2x}\,dx = c\int_0^{\infty} e^{-2x}\,dx$

$= c\left[-\frac{1}{2}e^{-2x}\Big|_0^{\infty}\right] = c\left[0 - (-1/2)\right] = c\,\frac{1}{2}$

$\Rightarrow c = 2$

$f(x) = 2e^{-2x}$

2) $P(1<X<2) = P(1 \leq X \leq 2) = \int_1^2 f(x)\,dx = \int_1^2 2e^{-2x}\,dx$

$= 2\left[-\frac{1}{2}e^{-2x}\Big|_1^2\right] = 2\left[-\frac{1}{2}e^{-4} + \frac{1}{2}e^{-2}\right]$

$= e^{-2} - e^{-4} \approx 0.117$

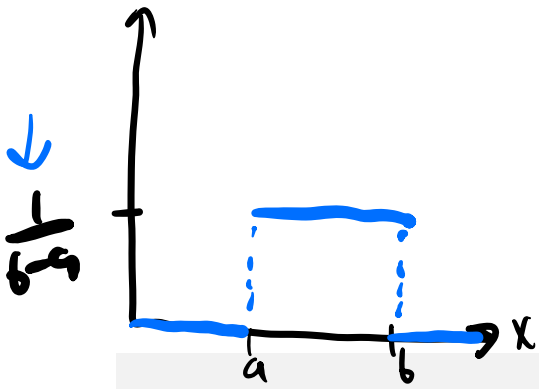
♢

4.1.2 A Couple Types of Continuous Random Variables

Definition 4.4 (Uniform)**.** The random variable X is **uniform** on $[a, b]$ if its pdf is

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{for } a \le x \le b \\ 0, & \text{otherwise} \end{cases}.$$

Example 4.4. Show that $\int_{-\infty}^{\infty} f(x)dx = 1$ for the uniform distribution.



$$\int_{-\infty}^{\infty} f(x)\,dx = \int_a^b \frac{1}{b-a}\,dx = \frac{x}{b-a}\Big|_a^b$$

$$= \frac{b}{b-a} - \frac{a}{b-a} = \frac{b-a}{b-a} = 1$$

♢

Definition 4.5 (Exponential)**.** The random variable X follows an **exponential** distribution with *rate* λ, if its pdf is

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{for } x \ge 0 \\ 0, & \text{otherwise} \end{cases}.$$

Example 4.5. Show that $\int_{-\infty}^{\infty} f(x)dx = 1$ for the exponential distribution.

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_0^{\infty} \lambda e^{-\lambda x}\,dx = \lambda \int_0^{\infty} e^{-\lambda x}\,dx = \lambda\left[-\frac{1}{\lambda} e^{-\lambda x}\Big|_0^{\infty}\right]$$

$$= \lambda\left[0 - \left(-\frac{1}{\lambda}\right)\right] = \lambda\left[\frac{1}{\lambda}\right] = 1$$

♢

4.2 Some Other Common Continuous Distributions

- Normal (Gaussian) distribution
- Gamma distribution
- The Beta distribution
- The student$-t$ distribution

- Cauchy distribution
- The χ^2 distribution

4.3 Cumulative distribution functions [DS 3.3]

Remember that the pmf is specific to discrete distributions and the pdf is specific to continuous distributions. We will now study the Cumulative Distribution Function (CDF), which can be defined for both types of random variables.

Definition 4.6 (CDF)**.** The cumulative distribution function (CDF) is the function $F(x)$, for all $x \in \mathbb{R}$, such that

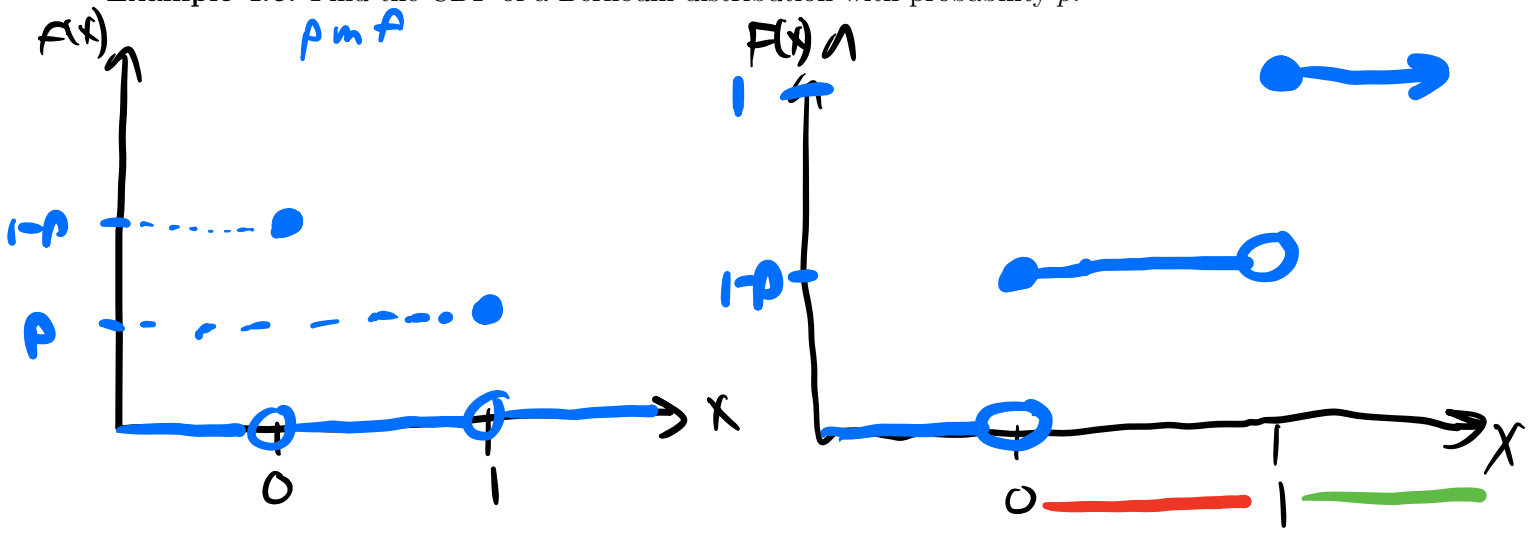
$$F(x) = \mathbb{P}(X \leq x).$$

Definition 4.7. The CDF of a discrete random variable, X is given by

$$F(x) = \mathbb{P}(X \leq a) = \sum_{x \leq a} f(x)dx,$$

where $f(x)$ is the probability mass function. If we know $F(x)$, we can also find $\mathbb{P}(X > a) = 1 - F(a)$.

Example 4.6. Find the CDF of a Bernoulli distribution with probability p.



$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1-p & \text{for } 0 \leq x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}$$

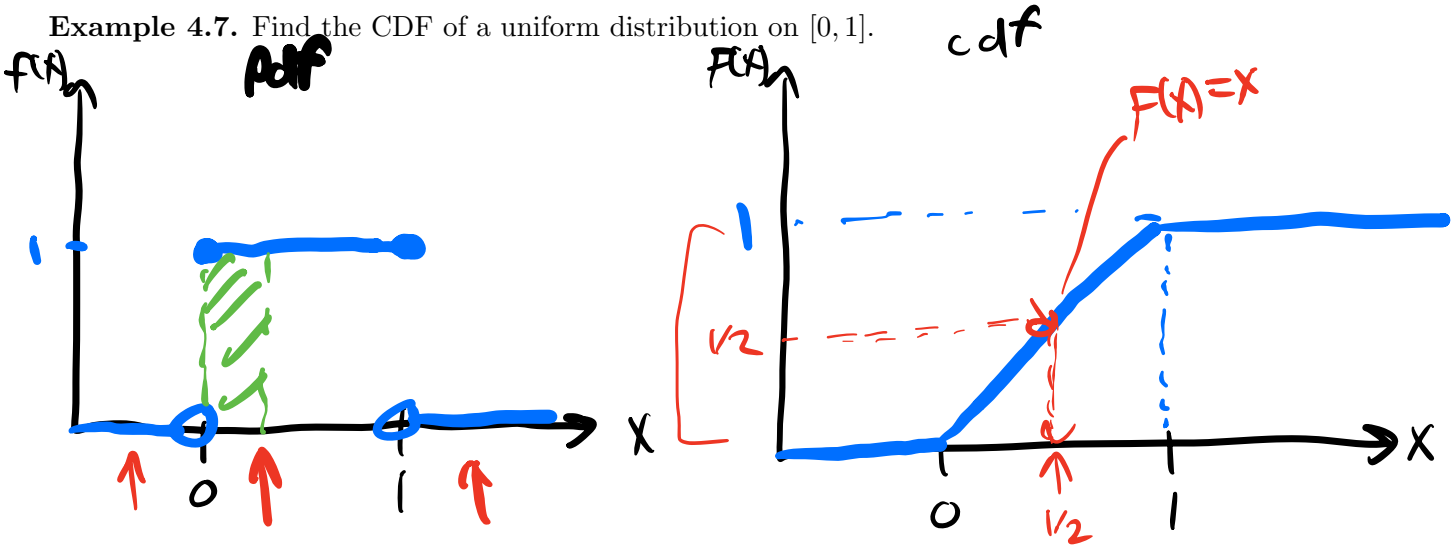
♢

Definition 4.8. The CDF of a continuous random variable, X is given by

$$F(x) = \mathbb{P}(-\infty < X \leq a) = \int_{-\infty}^{a} f(x)dx,$$

where $f(x)$ is the probability density function. If we know $F(x)$, we can also find $\mathbb{P}(X > a) = 1 - F(a)$.

Example 4.7. Find the CDF of a uniform distribution on $[0, 1]$.



$$F(x) = P(X \le a) = \int_0^x \frac{1}{b-a}\,dx = \frac{x}{b-a} = \frac{x}{1} = x, \qquad F(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \le x \le 1 \\ 1 & \text{for } x > 1 \end{cases}$$

♢

We have seen how to construct the CDF when we know the pdf, but we can also find the pdf when we know the CDF.

Theorem 4.1. *For a continuous random variable X,*

$$f(x) = \frac{d}{dx} F(x).$$

pdf

Proof. Use the fundamental theorem of calculus.

Suppose we have a cdf $F(x)$, then

$$\frac{d}{dx} F(x) = \frac{d}{dx} \int_{-\infty}^{x} f(t)\,dt = f(x)$$

↑ fundamental thm. of Calculus

□

Example 4.8. Suppose a random variable has the following CDF:

$$F(x) = 1 - e^{-5x}.$$

Find the corresponding pdf. What distribution is this?

$f(x) = \frac{d}{dx} F(x) = \frac{d}{dx} 1 - e^{-5x} = 5e^{-5x}$

This is the exponential dist. with $\lambda = 5$

♢

4.4 Properties of CDFs

- Non-decreasing: $F(x_1) \leq F(x_2)$ if $x_1 < x_2$
- $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$
- For every $x \in \mathbb{R}$, $\mathbb{P}(X > x) = 1 - F(x)$
- For all $x_1 < x_2$, $\mathbb{P}(x_1 < X \leq x_2) = F(x_2) - F(x_1)$

Definition 4.9 (Quantile)**.** Let X be a random variable with CDF $F(x)$. For each p strictly between 0 and 1, define $F^{-1}(p)$ to be the smallest value x such that $F(x) \geq p$. Then $F^{-1}(p)$ is called the p quantile of X or the $100p$ percentile of X. The function $F^{-1}(p)$ defined here on the open interval $(0, 1)$ is called the quantile function of X.

Definition 4.10 (~~Median~~ median/Quartile)**.** The 1/2 quantile or the 50th percentile of a distribution is called its median. The 1/4 quantile or 25th percentile is the lower quartile. The 3/4 quantile or 75th percentile is called the upper quartile.

Theorem 4.2. *When the random variable X is continuous, $F^{-1}(\cdot)$ is the inverse of $F(\cdot)$.*

Example 4.9. Find the quantile function for a uniform distribution on $[0, 1]$.

$F(x) = x$

Set $p = x$ and solve for x.

Then $F^{-1}(p) = p$.

The median is $F^{-1}(0.5) = 0.5$

♢

Example 4.10. Find the quantile function for an exponential distribution with rate $\lambda = 2$.

$F(x) = 1 - e^{-2x}$

Set $p = 1 - e^{-2x}$ and solve for x

$\Rightarrow p - 1 = -e^{-2x}$

$\Rightarrow 1 - p = e^{-2x}$

$\Rightarrow \log(1-p) = -2x$

♢

$\Rightarrow -\frac{\log(1-p)}{2} = x$

So $F^{-1}(p) = \frac{-\log(1-p)}{2}$

The median is $F^{-1}(0.5) = \frac{-\log(1-1/2)}{2} = \frac{\log(2)}{2}$